

System Analysis and Design

[Attendo]

**Smart Attendance &
Academic Monitoring
System**

Project Teamwork

No	Name	Role
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Introduction

Universities today face many challenges in effectively managing student attendance and monitoring academic performance. Traditional attendance methods, such as manual roll calls or paper-based records, are time consuming and prone to human errors. In many cases, students are not aware of their absence percentage until it becomes too late, which may result in academic warnings or even being denied from taking the final exam. To address these issues, this project proposes a Smart Attendance and Academic Monitoring System that aims to digitize and automate the attendance process while providing continuous monitoring of student participation.

Project Idea?

The system allows instructors to record attendance using modern techniques such as QR codes or mobile check-in within the university campus. In addition, it provides an early warning mechanism that alerts students when their absence percentage approaches the allowed limit. This helps students stay aware of their attendance status and avoid academic penalties. Furthermore, the system generates analytical reports that enable faculty members and administrators to study the relationship between attendance and academic performance. These reports can highlight courses with high absence rates and help identify potential academic issues. Through this system, students can easily track their attendance and receive warnings when necessary, instructors can manage attendance records efficiently, and administrators can access comprehensive statistics and reports. Ultimately, the system aims to improve academic monitoring, increase transparency, and support better decision-making within the university environment.

System Comparison Table

Aspect	Current System (Manual Attendance)	Proposed System (Smart Attendance & Academic Monitoring System)
<i>Attendance Recording</i>	Attendance is recorded manually by calling student names or using paper sheets.	Attendance is recorded digitally using QR code scanning or mobile check in.
<i>Accuracy</i>	Prone to human errors such as missing names or incorrect records.	More accurate and automated, reducing human errors.
<i>Student Awareness</i>	Students often do not know their absence percentage until it becomes critical.	Students can view their attendance percentage anytime through the system dashboard.
<i>Warning System</i>	No automatic warning system for excessive absences.	The system provides early warnings when a student approaches the absence limit.
<i>Reporting</i>	Reports are difficult to generate and usually require manual work.	The system generates automatic reports and statistics for instructors and administrators.
<i>Accessibility</i>	Attendance records are stored on paper and are difficult to access	Data is stored digitally and can be accessed easily by authorized users.

<i>Decision Support</i>	Limited data available for academic decision-making.	Provides Analytical reports to help understand relation between attendance and academic performance.
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System Request

Project sponsor	1. Mai Mohamed Ahmed 2. Nada Mohamed Hafez 3. Merna Safa Mogahed 4. Nada Khaled Elads 5. Mai Elsayed Abdelaziz
Business Need	• Digitalize the traditional manual attendance process to eliminate human errors. • Monitor student absenteeism patterns in real-time • Bridge the communication gap between students and academic advisors regarding attendance warnings. • Analyze the impact of attendance on overall academic performance.
Business Requirements	• Smart Attendance Recording: Support for QR Code check-in, Geofencing, or digital manual entry by instructors. • Early Warning System: Automated notifications when a student's absence exceeds the 25% threshold. • Academic Analytics: Dashboards for analyzing the correlation between attendance rates and midterm/final grades. • Excuse Management System: A portal for students to upload medical or official excuses for administrative review. • Multi-User Dashboards: Specialized views for Students, Doctors, Advisors, and Admins.
Business value	• Estimated project value: Reduction in administrative overhead costs by 30% through automation. • Improved Compliance: Ensuring 100% accuracy in enforcing university attendance regulations.

	Student Success: Potential 15% improvement in student grades due to early interventions and warnings •
Special issues or requirements	• The system must be launched before the start of the next academic semester (e.g., September 2026). • Must integrate with the existing University Database/LMS. • Compatibility with both Web browsers and Mobile apps (iOS & Android). • High data security to protect student records.

Feasibility Analysis

Technical feasibility	● Familiarity with Attendance Systems: The team has a strong understanding of academic workflows; existing similar solutions provide a clear roadmap. ● Familiarity with Technology: The team is proficient in web development (React/Angular) and backend (Node.js/Django). Advanced features like Geofencing and QR Generation might require extra research but are well-documented in modern APIs. ● Project Size & Timeline: The project is manageable for a team of 5 members within the semester timeframe
Economic feasibility	● Cost-Benefit Analysis: ROI: Estimated at 20% to 40% annually due to reduced paper waste and administrative time. Break-even Point: 8 to 12 months after implementation. ● Intangible Benefits: Enhanced reputation of the college as a "Smart Campus." Better student engagement and reduced dropout/failure rates. Faster decision-making for the administration based on real-time data.
Organizational feasibility	● Key Benefits for Users: For Students: Real-time tracking of their attendance status to avoid sudden "Denied" states.

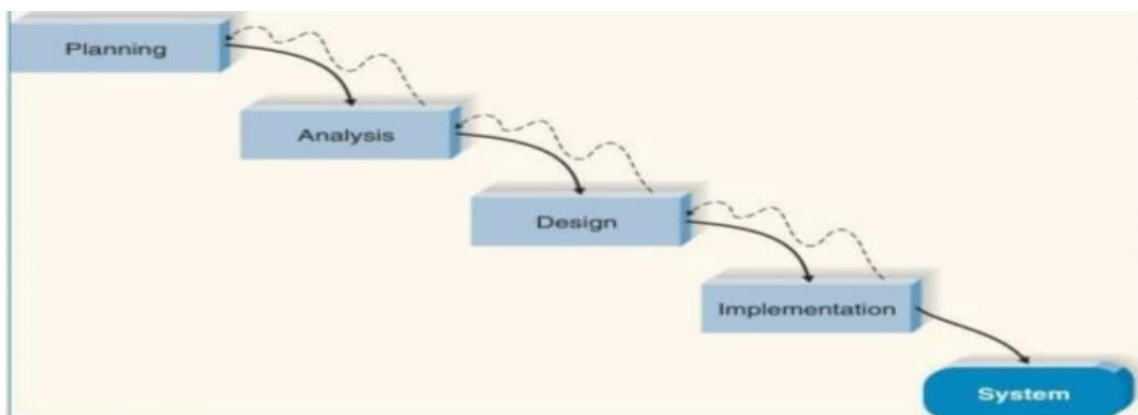
	For Doctors: Automated report generation and easy excuse management. For Admin: Full control and oversight of all departments' performance. ●Target User Groups: Undergraduate and Postgraduate students. Academic staff and Faculty Administration.
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System Methodology

Project Characteristics	Waterfall	Parallel	V-Model	Iterative	System Prototyping	Throwaway Prototyping	Agile Development
Requirements well defined	Excellent	Excellent	Excellent	Poor	Poor	Poor	Poor
Familiar technology	Excellent	Good	Good	Excellent	Good	Good	Good
Complex system	Good	Good	Good	Good	Poor	Excellent	Poor
Need for flexibility	Poor	Poor	Poor	Good	Excellent	Excellent	Excellent
User involvement	Poor	Poor	Poor	Good	Excellent	Good	Excellent
Time pressure	Poor	Good	Poor	Good	Good	Excellent	Excellent
Project visibility	Excellent	Good	Good	Poor	Poor	Poor	Poor

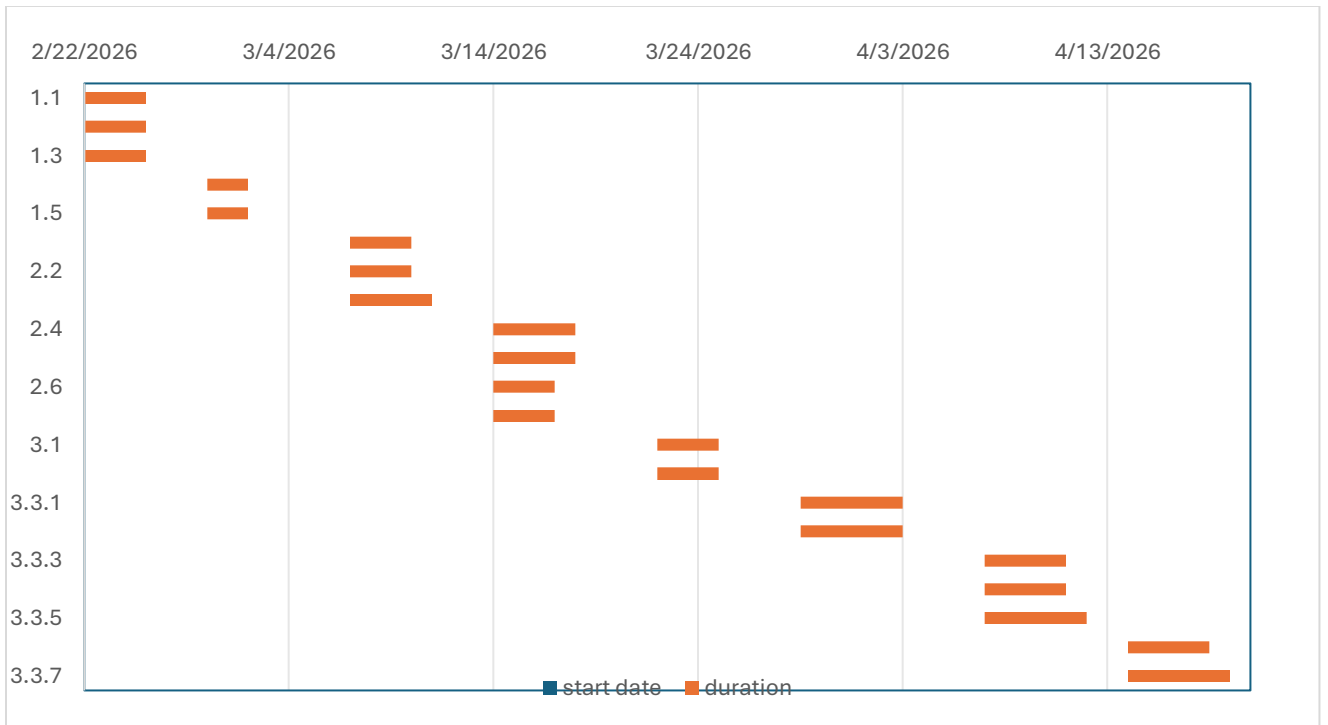
After analyzing the project requirements and constraints, we have selected the Waterfall Methodology for our "Smart Attendance & Academic Monitoring System." This choice is justified by the following reasons:

- **Clear User Requirements:** The functional requirements, such as QR code scanning, automated warnings, and reporting, are well-defined and stable from the beginning of the project.
 - **Fixed Scope:** Since this is a college-based project with a specific set of actors (Student, Doctor, Admin), we do not anticipate major changes to the system's core logic during development
 - **Structured Planning:** The Waterfall model provides a disciplined approach where each phase (Analysis, Design, Implementation) has specific deliverables, making it easier to manage within a 5-member team.
 - **Familiarity with Technology:** Our team has a clear understanding of the technologies needed (Mobile App & Database), which reduces the risks associated with uncertainty or constant changes.
 - **Time Schedule:** Given the semester timeline, the Waterfall approach allows us to set a firm completion date and ensure all documentation is ready before the final presentation
- Manage & Track: Each stage has a clear start and end point, allowing us to track our progress accurately against the project milestones



Project Plan

Phase No.	Phase Name	Task Id	Task Name	Assigned To	Estimated			Actual			Dependency	Status
					Duration	Start	Finish	Duration	Start	Finish		
1	Planning Phase	1.1	System Request	Mai Mohammed	3Days	27/01/2026	03/02/2026	2Days	28/01/2026	02/02/2026		Done
		1.2	Feasibility Analysis	Nada Khaled	2Days	03/02/2026	07/02/2026	3Days	04/02/2026	07/02/2026		Done
		1.3	Project Methodology	Mai Mohammed	3Days	07/02/2026	10/02/2026	4Days	08/02/2026	9/02/2026		Done
		1.4	Project Plan	Nada Hafez	4Days	10/02/2026	11/02/2026	5Days	11/02/2026	12/02/2026		Done
		1.5	Gantt Chart	Merna safaa	5Days	12/02/2026	18/02/2026	2Days	15/02/2026	18/02/2026	1.4.1	Done
2	Analysis Phase	2.1	Requirement	Mai ELsaed	2Days	18/02/2026	20/02/2026	4Days	17/02/2026	21/02/2026		Done
		2.2	Data Gathering	Mai Mohammed	1Days	20/02/2026	24/03/2026	3Days	19/02/2026	22/03/2026		Done
		2.3	Usecase Diagram	Merna safaa	4Days	24/03/2026	03/04/2026	1Days	24/03/2026	04/04/2026	2.1	Done
		2.4	Context Diagram	Mai ELsaed	6Days	03/04/2026	05/04/2026	5Days	04/04/2026	05/04/2026	2.3	Done
		2.5	Functional Decomposition	Merna safaa	4Days	05/04/2026	8/04/2026	6Days	05/04/2026	06/04/2026	2.3	Done
		2.6	DFD Level 0	Merna safaa	3Days	10/04/2026	15/04/2026	3Days	09/04/2026	12/04/2026	2.4 & 2.5	Done
		2.7	ER Diagram	Nada Hafez	5Days	18/04/2026	20/04/2026	3Days	15/04/2026	18/04/2026		Done
3	Design Phase	3.1	Alternative Matrix	Merna safaa	2Days	20/04/2026	22/04/2026	3Days	20/04/2026	22/04/2026		Done
		3.2	User Interface	Nada Hafez	3Days	22/04/2026	24/04/2026	3Days	22/04/2026	25/04/2026		Done
		3.3.1	Activity Diagram	Nada Hafez	4Days	24/04/2026	26/04/2026	2Days	26/04/2026	27/04/2026	2.3	Done
		3.3.2	Class Diagram	Merna safaa	2Days	26/04/2026	28/04/2026	2Days	26/04/2026	28/04/2026	2.7	Done
		3.3.3	Sequence Diagram	Mai ELsaed	3Days	28/04/2026	30/04/2026	2Days	28/04/2026	30/04/2026		Done
		3.3.4	Component Diagram	Mai Mohammed	1Days	30/04/2026	01/05/2026	3Days	30/04/2026	01/05/2026		Done
		3.3.5	Package Diagram	Nada Hafez	3Days	01/05/2026	03/05/2026	5Days	03/05/2026	04/05/2026		Done
		3.3.6	Object Diagram	Mai ELsaed	1Days	03/05/2026	04/05/2026	6Days	05/05/2026	07/05/2026	2.7	Done
		3.3.7	Deployment Diagram	Mai ELsaed	2Days	05/05/2026	07/05/2026	3Days	07/05/2026	010/05/2026		Done



Business Model

Key Partners • University administration • IT department • Academic staff and instructors • System developers and maintenance team	Key Activities • Developing the attendance system • Managing attendance records • Monitoring student attendance and absence • Generating reports and analytics	Value Propositions • Automated attendance tracking system • Reducing manual errors in attendance records • Early warning system for excessive absences • Providing real-time attendance insights for students and staff
Customer Relationships • Continuous system support • System notifications and alerts • User-friendly interface for students and staff	Customer Segments • Undergraduate students • Postgraduate students • Instructors and academic staff • University administration	Key Resources • System database • Web and mobile application • University network infrastructure • Development team

Channels	Cost Structure	Revenue Streams
• Web application • Mobile application • University portal or learning management system	• System development costs • System maintenance and updates • Server and database management	• University funding or budget allocation • Potential licensing for other universities

< Requirements Elicitation Techniques >

To gather the requirements for the Smart Attendance System, several elicitation techniques were used to understand the needs of students, instructors, and the administration.

1. Interviews

Interviews were conducted with instructors and administrative staff to understand the current attendance process and the problems they face, such as manual recording errors and time consumption.

2. Questionnaires

Questionnaires were distributed to students to collect their opinions about the current attendance system and their suggestions for improving it using a smart system.

3. Observation

The current attendance process was observed in classrooms to understand how attendance is taken manually and identify inefficiencies.

4. Document Analysis

Existing attendance sheets and reports were analyzed to understand the format and information required in the new system.

5. Prototyping

A simple prototype of the Smart Attendance System was presented to instructors to gather feedback and improve the system design.

<System Functional Requirements>

1. Check-In

1.1 The system allows students to record their attendance by scanning a QR code generated for each lecture session.

1.2 The QR code is created by the instructor at the beginning of the lecture.

1.3 The system verifies the identity of the student after scanning the QR code.

1.4 The system checks that the student is enrolled in the course.

1.5 The system records attendance with the lecture date and time.

2. Verify

2.1 This feature allows students to mark their attendance only when they are within the university campus area.

2.2 The system uses the device GPS to detect the student's location.

2.3 If the student is outside the allowed location range, the attendance request will be rejected.

2.4 If the location is verified, the system records the attendance in the database.

3. Warn

3.1 This feature monitors the attendance percentage of each student.

3.2 The system calculates the absence rate automatically.

3.3 If the absence rate exceeds the allowed threshold (for example 25%), the system generates a warning.

3.4 The warning notification is sent to the student.

3.5 The warning is also visible on the student's dashboard.

4. Submit

4.1 This feature allows students to submit absence excuses through the system.

4.2 Students can upload supporting documents such as medical reports.

4.3 The uploaded documents are stored in the system database.

4.4 The instructor or administrator reviews the excuse request.

4.5 The system updates the attendance record based on the decision.

5. View

5.1 This feature provides dashboards for different types of users.

5.2 Students can view their attendance percentage and warnings.

5.3 Instructors can view attendance records for their courses.

5.4 Administrators can monitor attendance statistics for all departments.

<System Nonfunctional Requirements>

1. Operational

1.1 The system runs on a web browser and mobile devices.

1.2 The system supports different user roles such as students, instructors, and administrators.

1.3 The system stores attendance records for each lecture session.

1.4 The system allows instructors to manage attendance sessions easily.

2. Performance

2.1 The system responds to user actions within a few seconds.

2.2 The system supports multiple users accessing the system at the same time.

2.3 The system processes attendance check-in quickly during lectures.

2.4 The system updates attendance records in real time.

3. Security

3.1 The system requires users to log in using their email and password.

3.2 Only authorized users can modify attendance records.

3.3 The system protects student data from unauthorized access.

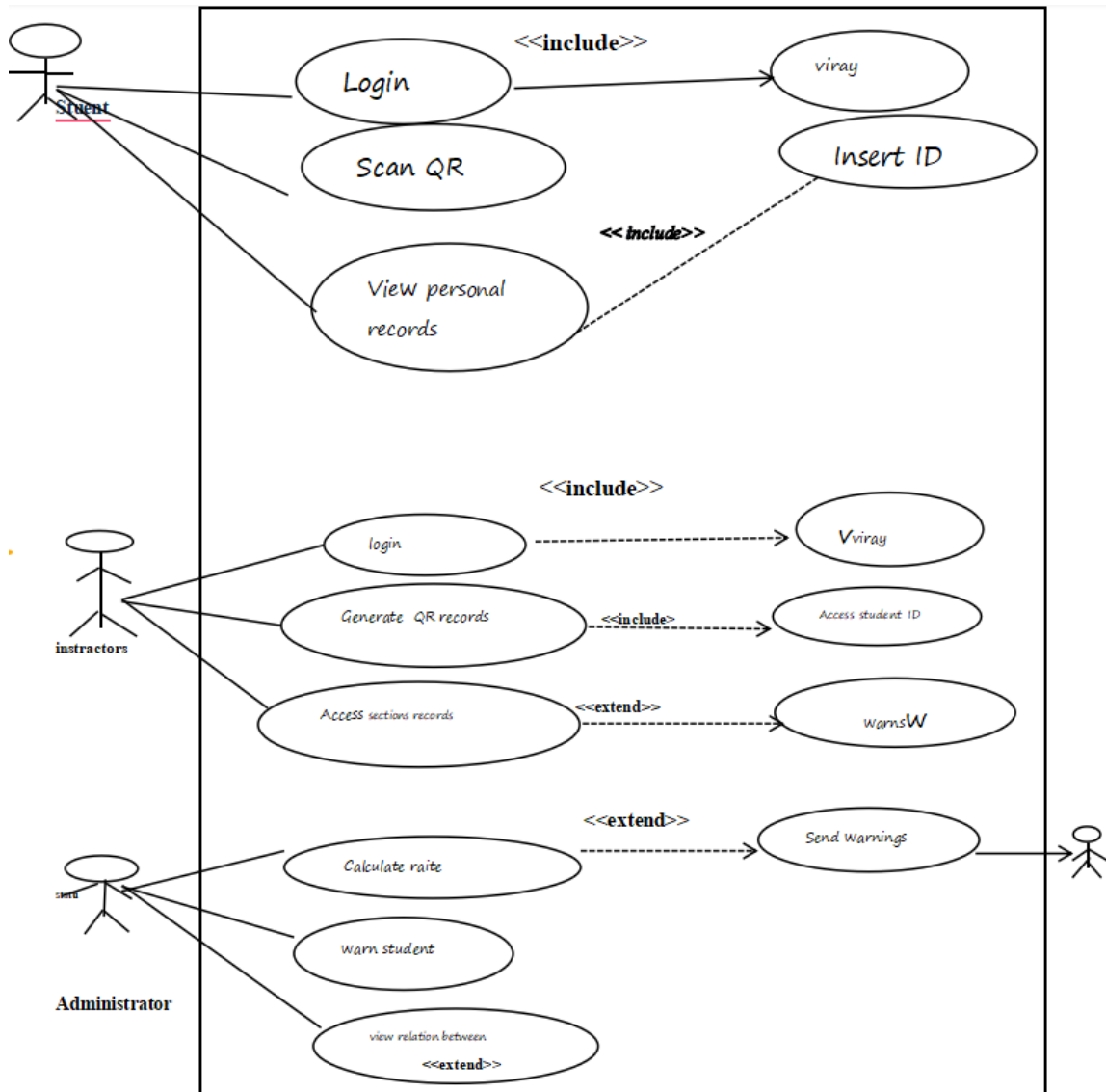
3.4 The system securely stores all attendance information.

4. Cultural & Political

4.1 The system supports both Arabic and English languages.

4.2 The system respects university policies and student privacy.

Usecase Figures



Tables for Usecases

Use Case 1: Scan QR (Attendance)

Field	Details
Use Case Name	Scan QR
ID	2
Priority	High
Actors	Students, Instructors
Type	Temporal / External
Description	Students use mobile app to scan QR code to record attendance automatically.
Trigger	Student scans QR code
Preconditions	Student logged in; Instructor generated active QR code
Normal Flow	1. Select Scan QR 2. Activate camera 3. Scan QR 4. Validate code & location 5. Record attendance 6. Show success message
Alternative Flow	If absence > 25% → warning triggered
Exceptions	Invalid QR code; No internet

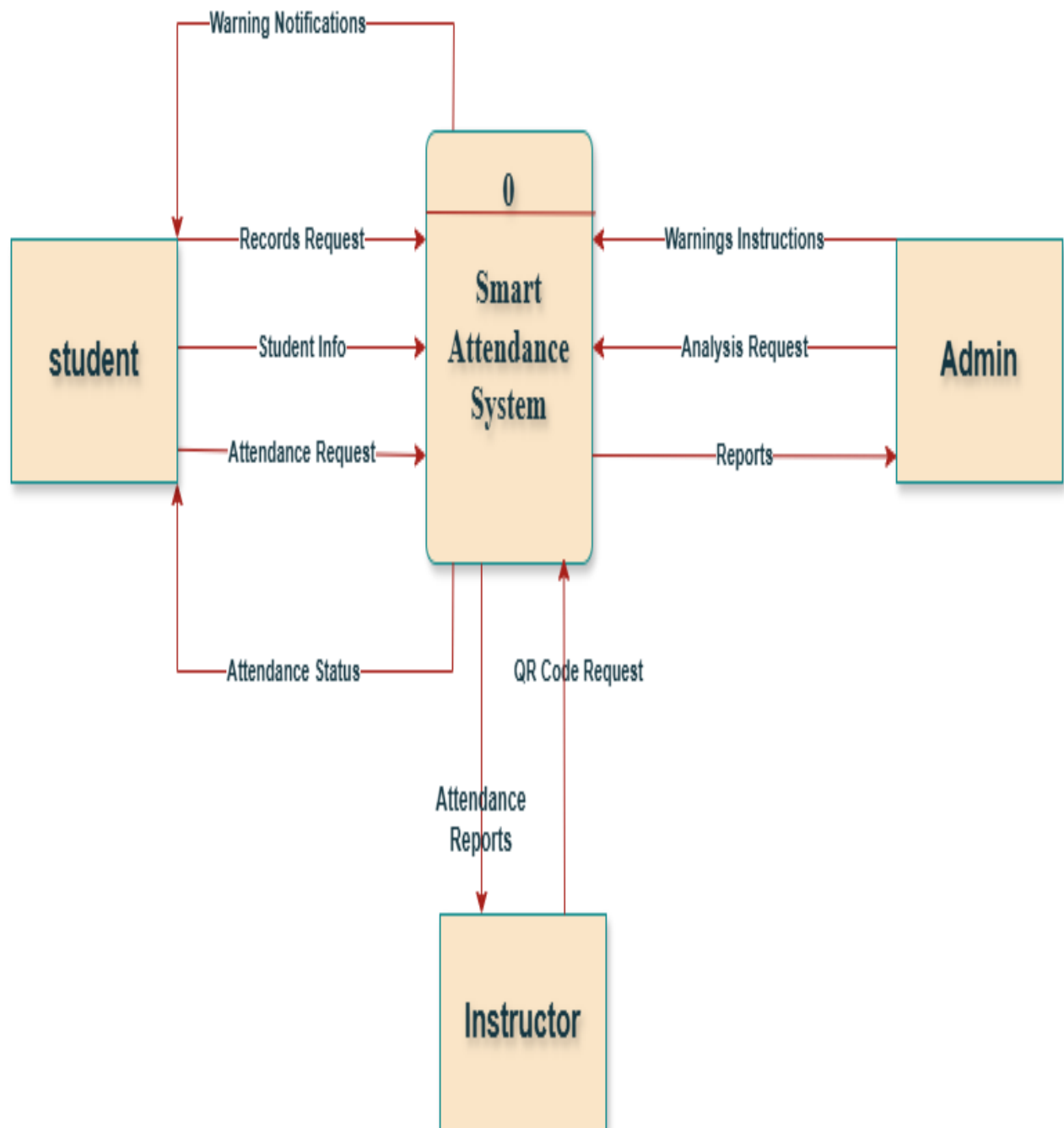
Postconditions	Attendance updated and percentage recalculated
Summary	QR from camera → Database attendance → Dashboard status

Use Case 2: Warn Student

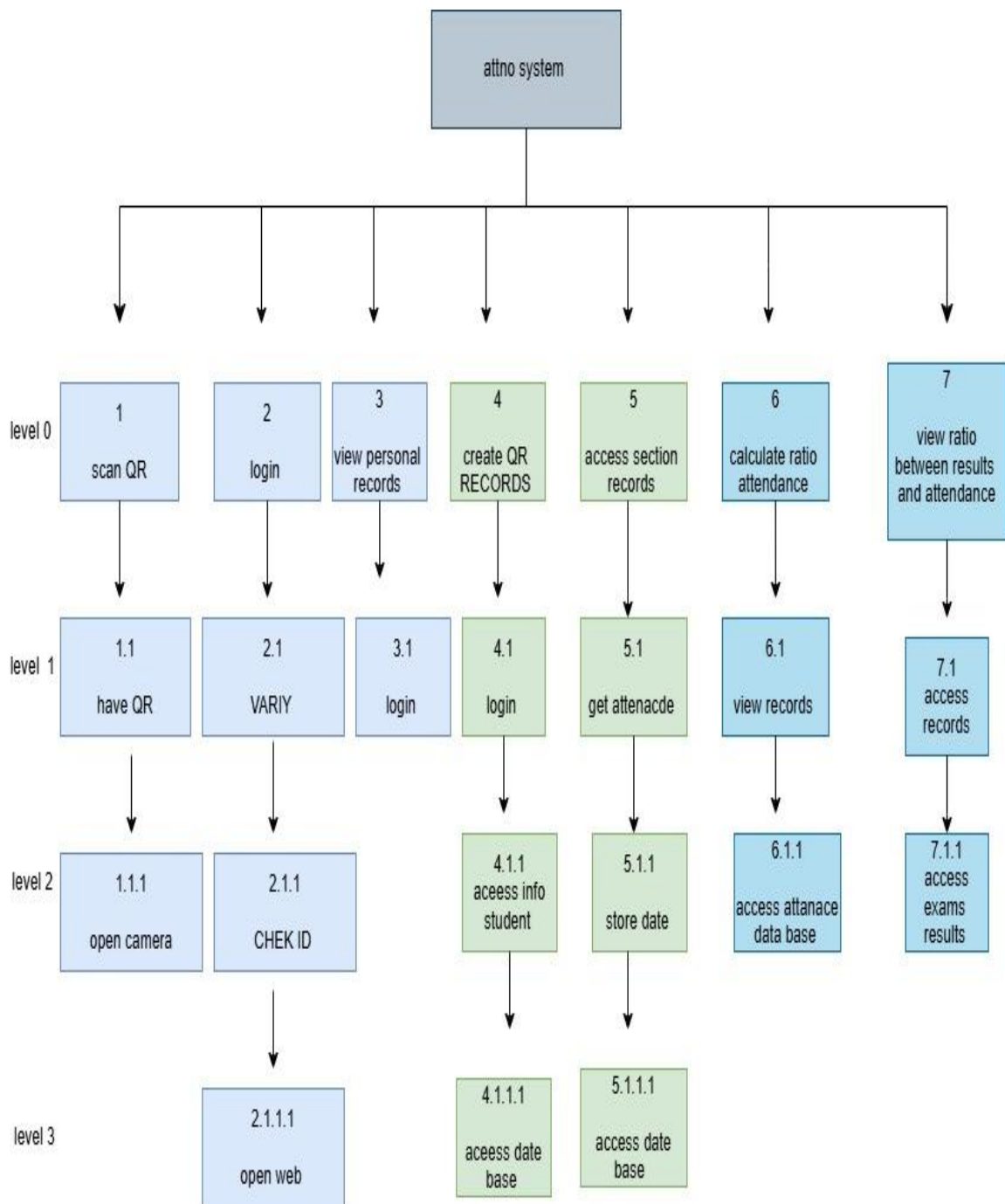
Field	Details
Use Case Name	Warn Student
ID	3
Priority	High
Actors	Administrator
Type	Temporal / External
Description	Admins warn students if absence > 25% affecting exams.
Trigger	System detects absence > 25%
Preconditions	Attendance updated; percentage calculated
Normal Flow	1. Check records 2. Identify students 3. Generate warning 4. Send notification 5. Log warning
Alternative Flow	Manual review before sending
Exceptions	DB failure; calculation error
Postconditions	Warning stored; report generated

Summary	Database → warning notification → student + admin report
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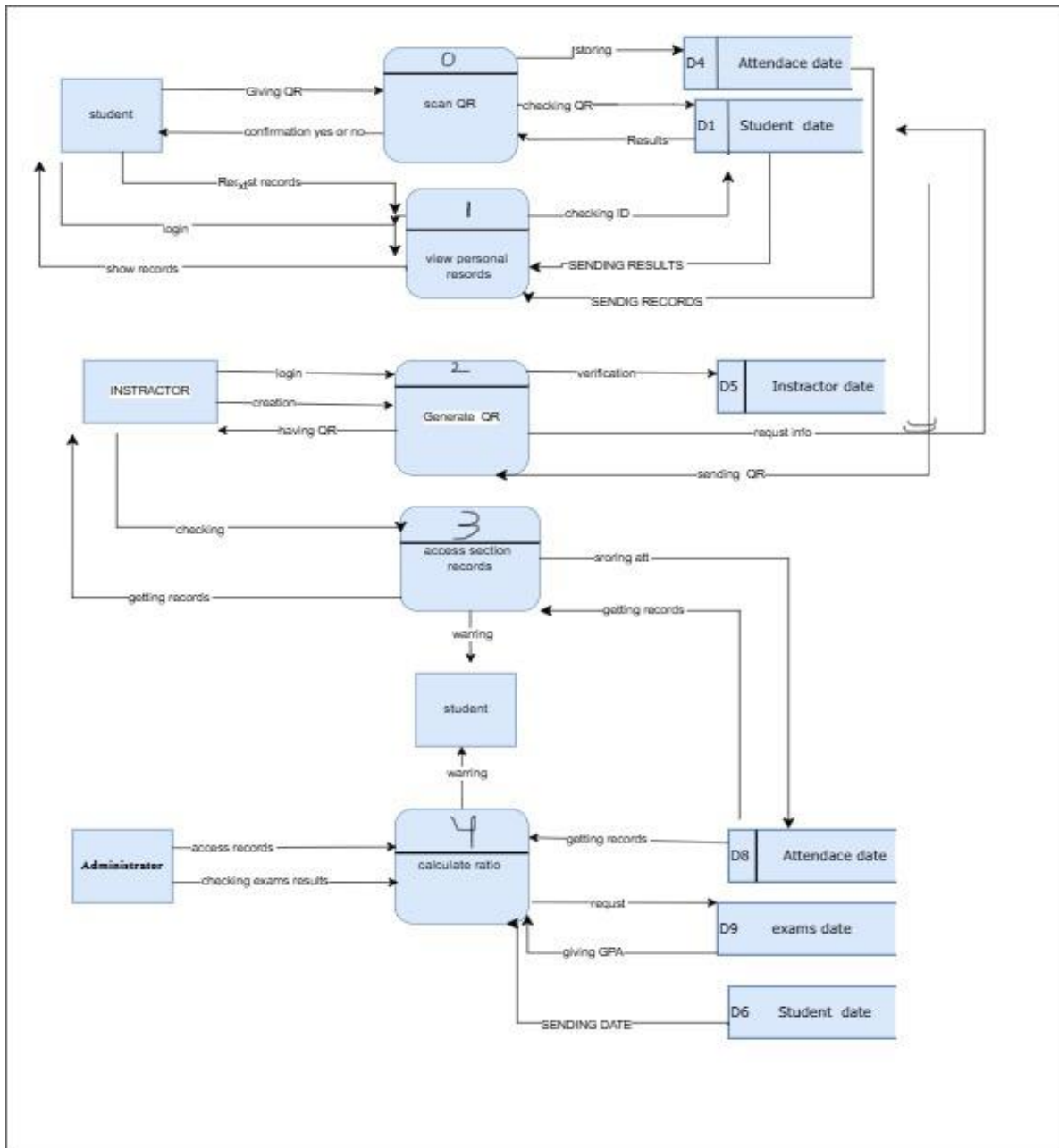
Context Diagram



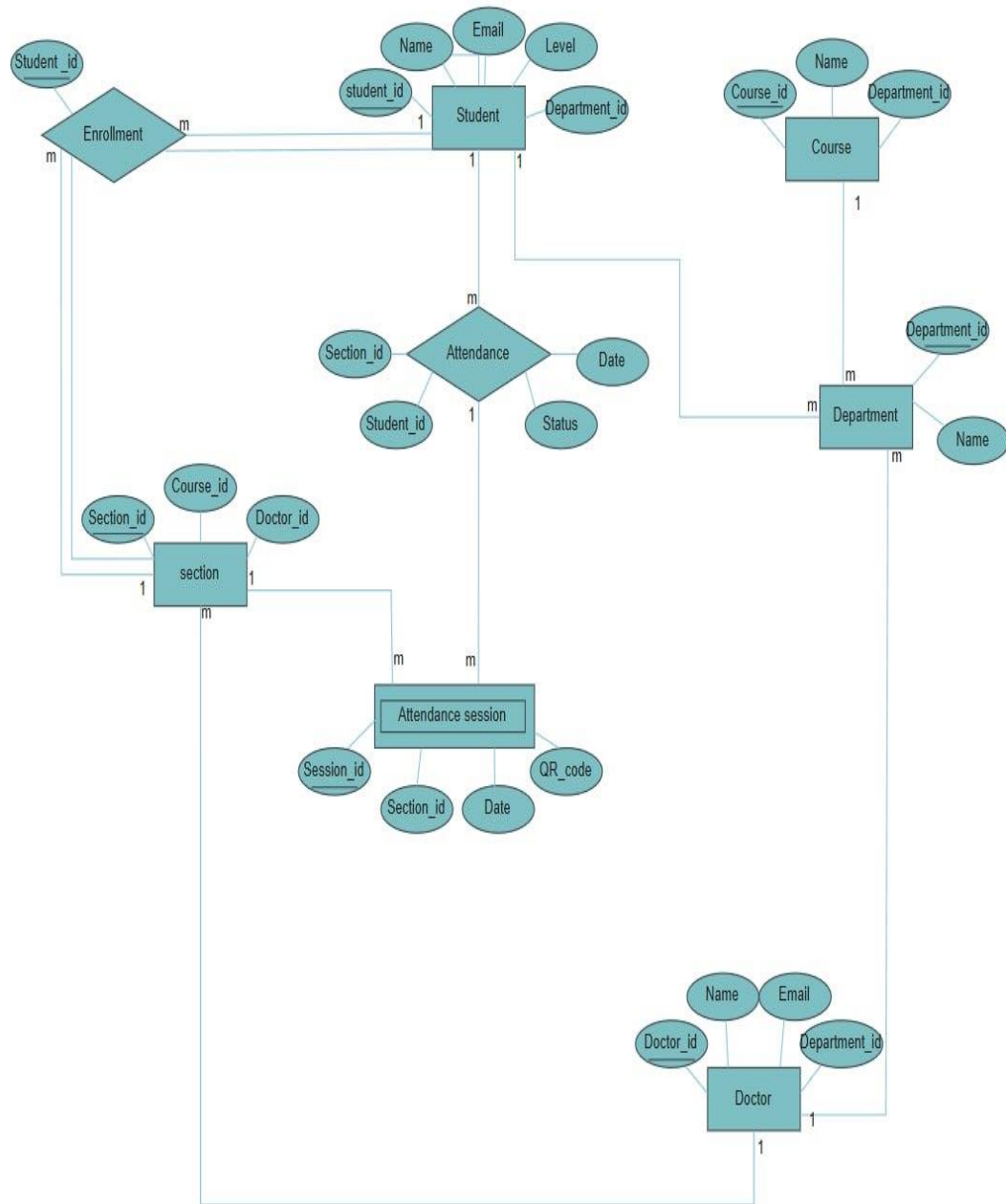
Functional decomposition



Dataflow Diagram with various Levels



ERD Diagram



System Acquisition Strategy

Project Characteristics	When to Use Custom Development	When to Use a Packaged System	When to Use Outsourcing
Business Need	The business need is unique and requires special features such as QR attendance, warnings, and analytics.	The business need is common and can be handled by existing attendance software.	The business need is important but the university prefers an external company to build it.
In-house Experience	The university team has technical knowledge and can manage development internally.	The university has functional experience but limited technical skills.	The university lacks technical experience and needs outside experts.
Project Skills	The required skills are available within the internal team.	The skills are not strategic and can be satisfied by packaged software.	Specialized development skills are needed from external professionals.
Project Management	The university has a skilled project manager to supervise internal development.	The project can be managed by coordinating with software vendors.	The university can manage contracts and supervise outsourced work.
Time Frame	The time frame is flexible and allows internal development.	The time frame is short and needs quick implementation.	The time frame is short but flexible enough to coordinate with vendors.
Cost	Higher initial cost but better long-term value and customization.	Lower initial cost with license or purchase fees.	Medium to high cost depending on vendor contract.
Flexibility	High flexibility for future changes and expansion.	Limited flexibility depending on package features.	Moderate flexibility based on outsourcing agreement.
Security & Control	Full control over student data and system security.	Limited control because system depends on vendor product.	Shared control with the outsourcing company.

The Alternative Matrix

Evaluation Criteria	Weight	Flutter App	Score	Weighted	Web App (React)	Score	Weighted	Low-Code Tool	Score	Weighted
Technical Issue										
Develops in-house skills	20	Team gains mobile dev skills	5	100	Team familiar with React	4	80	No coding needed	3	60
Integration with DB/API	15	Highly flexible	5	75	Highly flexible	5	75	Limited	2	30
Team experience	10	Limited experience	3	30	Good experience	4	40	No experience needed	5	50
Economic Issue										
Cost	25	Moderate	4	100	Low	5	125	Very low	5	125
Development time	10	3–4 months	3	30	Faster	4	40	Very fast	5	50
Organizational Issues										
Market readiness	10	Not common	3	30	Widely used	4	40	Very common	5	50

Customization	10	Fully customizable	5	50	Highly customizable	4	40	Limited	3	30
TOTAL	100			415			440			395

User Interface Design

(System screens, menus and options design)



App link: <https://fantastic-marigold-cd641c.netlify.app/>